ParkGuide AI: A Citation-Grounded Retrieval-Augmented Chatbot for U.S. National Park Visitor Information

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MSAI 633, University of the Cumberlands | October 2026 | Repository: github.com/cleveee/parkguide-ai | Demo: parkguide-ai-dh24.vercel.app

ParkGuide AI answers visitor questions about Yosemite, Yellowstone, and Grand Canyon National Parks using only saved pages from the National Park Service (NPS) website. Every delivered claim must cite a saved passage, and the system refuses or asks for clarification when evidence is missing. **Status note:** all results below are development diagnostics.

1. System Architecture

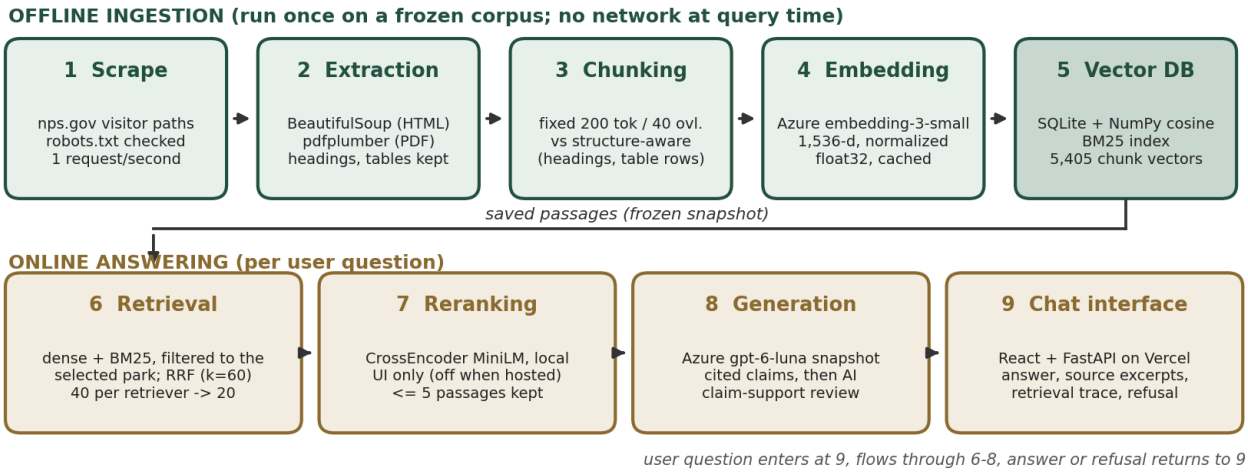


Figure 1. ParkGuide AI pipeline: offline ingestion (stages 1-5) feeds online answering (stages 6-9).

Offline. The crawler stays inside the visitor-planning paths of www.nps.gov for the three parks. It fetches and saves robots.txt first, shares one global limiter of one request per second across fetches, redirects, and retries, rechecks every redirect target, and writes raw bytes with a SHA-256 hash before any parsing. Extraction uses BeautifulSoup and lxml for HTML (navigation removed, heading hierarchy and tables kept, merged cells expanded) and pdfplumber for PDFs (page numbers kept). The frozen corpus has 155 documents: 150 HTML pages and 5 PDFs, of which 45 contain tables (Yosemite 56, Yellowstone 40, Grand Canyon 59). Chunks (Section 3) are embedded with Azure text-embedding-3-small into unit-normalized 1,536-dimension float32 vectors and stored with the documents in one SQLite file, searched by exact NumPy cosine similarity alongside a BM25 index.

Online. The user must pick one park first; every search is filtered to it and the history clears when the park changes. Dense and BM25 results are fused, optionally reranked, passed to the generator, and the checked answer is shown in a React and FastAPI interface hosted on Vercel with an inspector for evidence, retrieval scores, and token cost. The delivered stack differs from the proposal (Chroma, MiniLM, GPT-4.1 mini, and Streamlit were replaced by SQLite/NumPy, Azure embeddings, Azure Luna, and React/FastAPI) as part of a simplification onto one Azure resource and a read-only hosted profile.

2. Retrieval & Generation Design

Models and retrieval. Embeddings use text-embedding-3-small. Answers and claim review use the Azure gpt-6-luna deployment (returned snapshot gpt-6-luna-2026-09-22) at temperature 0, strict JSON output, storage disabled, a 1,000-token completion ceiling, and no SDK retries, under a $5 spend cap. Retrieval takes 40 ranks from dense search and from BM25, merges them by reciprocal rank fusion (k=60) into 20 candidates, and keeps at most five passages. Metadata filters apply before scoring and are never relaxed silently, and a keyword gate requires BM25 candidates to contain at least one concrete term from the current question.

Prompt and output design. The generator does not write free text. It returns an action (answer, clarify, or refuse) and a list of factual claims, each tied to the IDs of retrieved passages, and the visible answer is rebuilt from those claims. We chose this contract so every sentence is auditable: unknown or duplicate IDs, uncited claims, and truncated or malformed output fail validation. Retrieved text is treated as untrusted evidence, earlier assistant turns can resolve references but never count as evidence, and a caller-supplied system role is rejected.

Citations and the 25-word limit. Evidence IDs resolve to stored URLs (generated URLs are not allowed), and the interface shows numbered citations with the saved excerpt and a link to the current NPS page. A second Luna call receives only the cited passages and must return a decision for every claim with an exact witness from the saved text; unsupported claims are withheld. This is an AI entailment check, not a guarantee of accuracy. A quotation guard runs during claim checking and blocks answers that copy more than 25 consecutive source words, so answers must paraphrase while still citing.

Insufficient context. Four layers apply. (1) A scope gate refuses wrong-park, live-status, private-booking, and personal hiking-safety requests. (2) If filters leave a short or empty candidate pool, the shortfall is recorded and the user is asked to clarify instead of widening the search. (3) Claim review withholds anything the cited text does not support, and if nothing survives the system returns a refusal with related NPS links. (4) The interface states that pages are a snapshot, and safety-sensitive questions receive a limitation and a referral to official sources. We set no numeric answerability threshold: the decision rests on reviewed claim support rather than on an uncalibrated score.

3. Chunking & Re-Chunk Experiment

Strategies. The fixed baseline uses 200-token windows (cl100k_base tokenizer) with 40-token overlap and produced 1,698 chunks. The structure-aware strategy packs adjacent paragraphs under the same heading, splits oversized paragraphs, keeps each table separate, and repeats table headers when a table spans chunks. It produced 3,707 chunks and carries the heading path into every chunk.

ID scheme. doc_id is the SHA-256 of the canonical URL. A chunk ID is a hash of the content revision, the cleaner and chunker configuration, the source character span, and the chunk text, so identical content and configuration always yield the same ID. Gold evidence points to source spans rather than chunk IDs, so re-chunking does not invalidate labels.

Re-chunking without re-crawling. Replacement rows are built from the saved normalized blocks (data/documents.jsonl) with networking disabled, validated, and activated atomically in a single SQLite transaction, after which obsolete derived rows are deleted. The offline proof on a copy of the database reported zero network calls, zero stale chunk IDs, and zero orphaned vectors. The database holds 5,405 vectors (1,698 + 3,707) across the two builds with zero foreign-key errors. The same mechanism let us re-extract from unchanged raw bytes when fixing table parsing (Section 6). Retrieval effects of the two strategies appear in Table 1.

4. Version Control & Team Collaboration

The code, saved corpus, and run evidence are published at github.com/cleveee/parkguide-ai. Our proposal planned feature branches, a protected main, issues with owners, and at least one teammate approval per pull request, with four roles (ingestion; chunking and retrieval; generation and interface; evaluation and ethics) that each contribute to a second stage and review another role's changes. The repository records 11 commits on main and 4 closed pull requests. Implementation followed the repository's AGENTS.md rules: each AI coding agent owned specific files, a separate agent performed independent review, one integrating agent merged changes and owned Git and deployment, and credentials stayed in a git-ignored .env file. Human verification of the gold set was deliberately kept out of agent hands.

Member	Contribution
Pratik Bhavsar	Project Manager/Coordinator: Created the shared Drive folder and Microsoft Teams team, authored and organized project documents, coordinated between the engineering and data-science members, supported evaluation items, and prepared the Research Paper.
Adithya S. Gummadi	**Project Manager/Coordinator:** Prepared and organized the project presentation, summarized the problem statement and technical architecture, and developed speaking scripts explaining the source corpus and question-to-answer pipeline. Worked alongside Pratik Bhavsar to support team coordination and presentation preparation.
Danail Banaov	**Data Scientist** my contribution to PARKguideAI focused on human checking and evaluation. I reviewed the chatbot's answers against the saved NPS sources, recorded observations in GoldenQ, and identified issues such as incomplete answers, unnecessary refusals, and questions outside the chatbot's scope. This work helped assess the system's reliability and identify areas for improvement.
Naaman Ondega	**Data Scientist** my contribution to the ParkGuideAI was to help with the evaluating the Golden Set questions with Danail against the Knowledge Base and recording observation in the human verification process. I also assisted with the evaluation metrics
Robin Cleve	**Engineer:** Built the RAG pipeline, integrated Azure models and SQLite, and developed the React/FastAPI web app. Added park selection, source previews, and retrieval inspectors. Improved citation safeguards and conversation reliability, deployed the application on Vercel, and supported testing and technical documentation.
Surya Kolavennu	**Engineer:** Contributed to the design and implementation of ParkGuide AI, focusing on the Retrieval-Augmented Generation (RAG) architecture, document ingestion and chunking, and integration of AI-powered retrieval components. Supported the use of Azure OpenAI embeddings, SQLite-based vector storage, and hybrid retrieval techniques to improve the relevance of information retrieved from National Park Service documents. Assisted with system testing, troubleshooting, and reviewing grounding and citation safeguards to support reliable, evidence-based chatbot responses.
Vishal Talla	**Engineer: C**ontributed to building Park Guide AI. Additionally implemented persistent chat history and support for multiple saved conversations. I also added up to three relevant follow-up questions grounded in the available park data. I validated these features through automated tests and browser checks and managed the changes through Git feature branches and pull requests.

5. Ethics Considerations

Risks and mitigations implemented. (a) Stale or unsafe visitor advice: every answer is labeled as based on a saved snapshot, source age is shown, live closures and availability are refused, and personal safety judgments go to rangers. (b)

False confidence from citations: citations link to saved excerpts, an AI support review withholds unsupported claims, and the interface says answers rely on an AI source review. (c) Crawling harm: robots rules, a one-request-per-second limit, an allowlist of NPS paths, and exclusion of login, profile, and staff-directory pages. (d) Personal data: email and phone patterns are redacted, chat history lives only in browser-tab memory, API storage is disabled, and Azure processing is disclosed. (e) Prompt injection: retrieved text is untrusted and cannot supply a system role. (f) Cost and AI-use transparency: a $5 cap, and disclosure that OpenAI Codex-family agents drafted and implemented much of the code and that five AI models plus a separate assessor produced the reference labels. No agent certified human review.

Remaining limitations. Human verification of gold labels is pending, and the AI reviewer is uncalibrated: it can over-refuse and once approved an unsupported map-based absence claim. PII redaction is pattern-based, not comprehensive. The crawler contact is a placeholder address (parkguide-poc@example.com) approved as a course exception. Rotated PDF column labels and possible column interleaving need manual checks. The hosted app omits the reranker, and its budget ledger is per instance rather than global. **With more time** we would verify the gold set with human reviewers first, calibrate the AI reviewer against their labels, use a monitored crawler contact, and set up branch protection and PR review from day one.

6. Challenges & Solutions

Table and PDF extraction. NPS wraps pages in a layout table that was mistaken for data, Yellowstone tables use merged multi-row headers, and a Grand Canyon PDF has rotated column labels in reversed glyph order. We treated layout wrappers as layout, combined multi-row headers into single labels, re-extracted from saved raw bytes without re-crawling, and flagged the rotated labels for manual comparison with the raw PDF.

Over-refusal and table arithmetic. Early numeric and quotation guards withheld valid paraphrases and table calculations. We corrected the prompt and guards, now compute differences from cited cells (for example, 3 - 1.5 = 1.5 miles) and show the calculation in the trace, and require each claim to cite its actual table row. On the diagnostic items, the selected configuration then produced 12/12 clearly correct answers and 16/18 clearly correct dialogue turns, with two uncertain (AI-reviewed).

History aliasing bug. Later turns were being appended to earlier saved conversation histories, contaminating traces through stage 6. We snapshot each history before use; stage 7 histories hold exactly 0, 2, or 4 preceding messages with recorded code hashes. **Reviewer reliability:** because the AI reviewer sometimes approved an unsupported claim, it sees only the cited passages and is documented as a heuristic that needs human calibration.

7. Evaluation

Design. The repository contains two distinct evaluation tracks. The original course gold drafts were sealed before initial live answers, with a planned human-reviewed 24/16 development/test split (seed 633). That reviewed split and final-test benchmark remain pending. The later gold-matrix-v1 study uses revised, product-aligned candidates after development exposure; it does not inherit the original drafts' pre-output or held-out status.

Earlier retrieval experiment. Table 1 preserves the earlier controlled comparison on 12 diagnostic items (8 with source evidence), using GPT-4.1 mini rather than the later Luna configuration. B0 is closed book; B1 and B2 use fixed chunks; C1, C2, and B3 are structure-aware retrieval ablations. Its paired intervals use 10,000 bootstrap resamples (seed 633).

Variant	Retrieval / chunking	MRR	Action agreement	Lexical pass
B0	No retrieval (closed book)	n/a	12/12	2/12
B1	BM25, fixed	0.875	9/12	4/12
B2	Dense, fixed	0.550	8/12	3/12
C1	Dense, structure-aware	0.854	10/12	6/12
C2	Hybrid, structure-aware	0.854	11/12	5/12
B3 (earlier run)	C2 + CrossEncoder reranker	0.781	9/12	5/12

Table 1. Earlier controlled retrieval comparison. Paired MRR changes: B1 to B2 -0.325 [-0.563, -0.100]; B2 to C1 +0.304 [+0.083, +0.513]; C2 to B3 -0.073 [-0.219, +0.042]. Top-5 evidence coverage was 8/8 for every retrieval variant.

Structure-aware chunking improved dense MRR by 0.304 in this exploratory comparison. Reranking showed no established gain and added 340 ms mean latency (95% paired interval 305 to 365 ms). These eight evidence-bearing items do not establish a general advantage, and this earlier baseline comparison is separate from the 40-question study below.

7.1 Saved gold matrix outcomes

Protocol. On 2026-10-10, local chat.answer at commit 4d3faaf completed 40/40 requests with 0 execution errors. Each request selected one park and began with empty history. The run used hybrid retrieval, no reranker, and Azure Luna (gpt-6-luna-2026-09-22), including generation and claim checks. Settings match the hosted profile; execution and latency were local, without Vercel/API/browser measurement.

Five independent AI labeling configurations produced the 75-fact consensus reference for 40 questions. Candidate-category lineage is 15 factual, 6 multi-hop, 4 comparative, 4 temporal, 8 unanswerable, and 3 ambiguous/adversarial; five of the six multi-hop references use multiple pages, while PG016 resolves from one. A separate gpt-5.6-sol/high assessor graded the saved responses. Human verification and assessor calibration remain pending. No new chatbot or grading calls were made for this report revision.

Measurement	Saved result	Unit
Correct action	37/40 (92.5%)	Questions
Full rubric pass, AI assessed	21/40 (52.5%)	Questions
Required facts delivered, AI assessed	55/75 (73.3%)	Required facts
Behavior requirements satisfied, AI assessed	30/42 (71.4%)	Requirements
Delivered claims supported, AI assessed	40/40 (100.0%)	Claims in 27 answers
Delivered claims with valid citation identity	40/40 (100.0%)	Delivered claims
MRR / nDCG at 10	0.794 / 0.769	30 answerable cases
Recorded latency p50 / p95	4.14 s / 34.92 s	40 local requests

Table 2. Saved local end-to-end backend RAG evaluation. A full pass requires the correct action, all required facts and behaviors, supported delivered claims, and no additional answer issue; uncertain judgments do not pass.

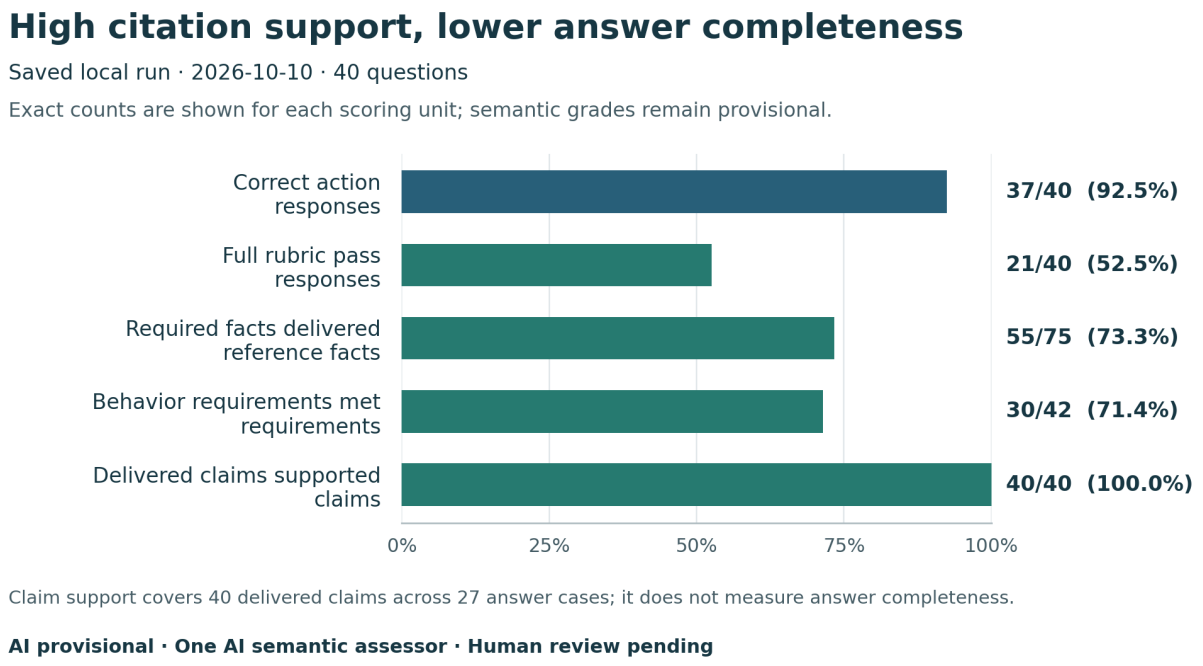


Figure 2. High action agreement and cited-claim support coexist with incomplete answers. Denominators differ by metric and are printed beside each bar. All semantic results are provisional AI assessments.

These descriptive rates are not population accuracy estimates. Facts and claims cluster within questions, sources overlap, and the questions have prior development exposure. Independent-observation confidence intervals would omit those dependencies and uncertainty in both the reference and assessor.

7.2 Evidence coverage and delivered information

Evidence coverage asks whether retrieved source spans cover a declared evidence group for a required fact. Delivered-fact coverage asks whether the answer conveys that fact according to the AI assessor. Figure 3 uses the same 75-fact denominator for each rate, while preserving the distinction between these measurements.

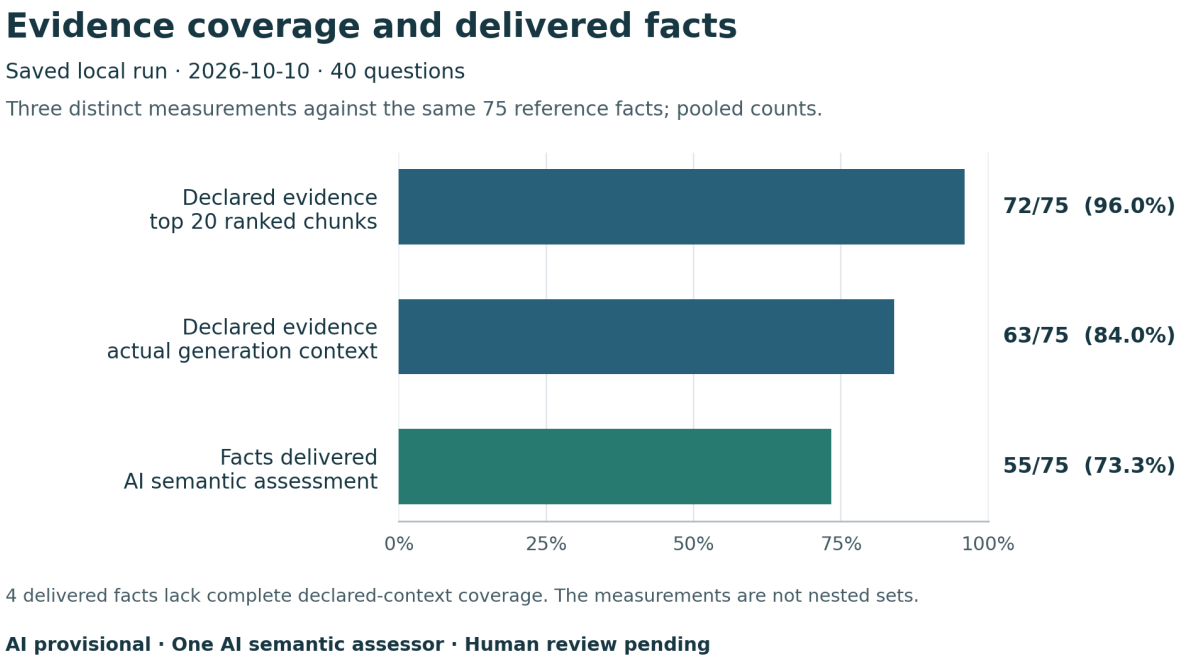


Figure 3. Separate pooled measurements of declared evidence and delivered facts. These bars are not a nested funnel and their differences do not measure causal losses at successive stages.

Evidence measure	Pooled facts	Mean per question
Top 5 ranked candidates	63/75 (84.0%)	84.2%
Top 20 ranked candidates	72/75 (96.0%)	96.7%
Actual generation contexts	63/75 (84.0%)	84.2%

Table 3. Pooled rates count 75 required facts; per-question means give equal weight to each of the 30 answerable cases. Top-k ranks precede context pruning. The two aggregations answer different questions.

Interpretation. Four facts judged delivered lack complete declared evidence-group coverage in the actual contexts: M008 F1, M010 F1/F2, and M034 F1. Declared alternatives are not exhaustive human relevance judgments, so this mismatch alone does not demonstrate unsupported content. A per-fact review of source spans and semantic labels is needed before attributing errors to context selection or generation.

Chunk retrieval. At rank 5, 35/150 returned chunks match declared relevance and 35/44 known relevant chunks are recovered: micro precision 23.3%, recall 79.5%, F1 36.1%. At rank 20 the corresponding counts are 43/600 and 43/44, yielding 7.2%, 97.7%, and 13.4%. These chunk metrics differ from required-fact coverage; the same fact can have several supporting chunks.

7.3 Action errors and incomplete responses

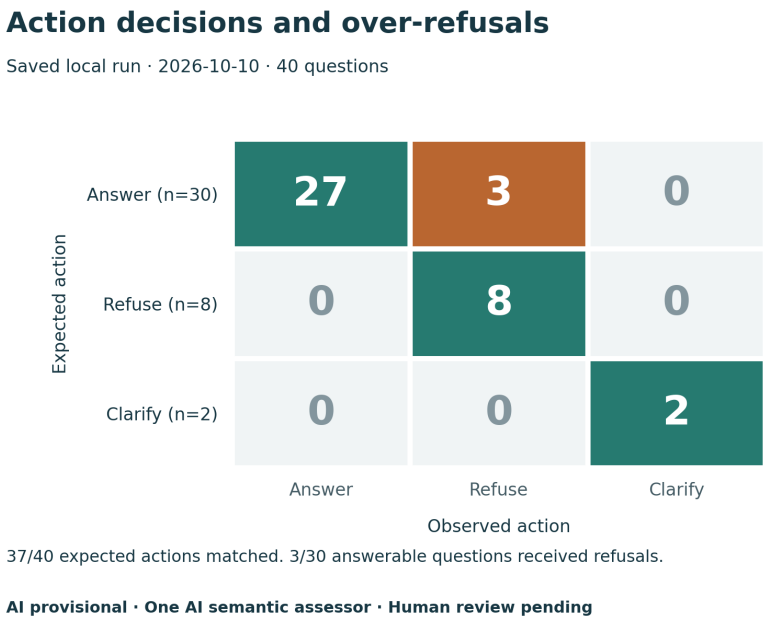


Figure 4. Expected versus observed actions for all 40 questions. The three disagreements are over-refusals of answerable questions. This action classification does not measure all requirements of the response.

Answer recall is 90.0%; refusal precision is 72.7%. All eight expected refusals and both expected clarifications received their expected action. Full rubric compliance is lower because missing facts, referrals, clarifying behavior, and caveats also matter. Action micro precision, recall, and F1 are each 92.5%; macro F1 is 93.0%.

Primary issue	Cases	Case IDs
Missing required facts	8	M004, M005, M012, M014, M025, M034, M037, M039
Over-refusal	3	M024, M027, M036
Missing or misdirected referral	3	M011, M020, M029
Missing clarifying behavior	2	M007, M026
Missing snapshot caveat	2	M010, M013
Inadequate source-limit explanation	1	M031

Table 4. The 19 unsuccessful cases, grouped once each by their principal recorded issue. Case-level reasons and all 40 responses remain available in the linked artifacts.

M036 triggered number_without_source_witness, so its missing answer includes a numeric-guard refusal. M026 and M034 have uncertain semantic labels, which remain failures under the scoring rule. No unsuccessful case was removed, relabeled, or retried for this revision.

8. Future Work

Human-review the original course reference labels, freeze the planned 24/16 development/test split, and run the B0-B3 final comparison once under its evaluation gates. Separately, review the product-aligned matrix and calibrate its AI assessor against human grades. Use development cases to improve completeness and reduce over-refusal, then evaluate the hosted API/browser path and multi-turn conversations. Verify table/PDF extraction and replace the placeholder crawler contact. None of these pending measurements is established by offline replay.

Reproducibility. The repository preserves the original paper, saved responses, exact metric counts, and reproducible PNG/SVG charts. Run scripts/evaluate.py --check to reproduce scores without provider calls; the figures and manuscript have separate offline check commands documented in the evaluation workflow.

Evaluation report and methods

All 40 question results and recorded failures

Figure data, source hashes, and reproduction workflow